Decoupling-Oriented Coordination of Paralleled VSGs With Multi-Agent Reinforcement Learning

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***Abstract*—Paralleled virtual synchronous generators (VSGs) must provide aggregate frequency support while suppressing differential motion among units. If a learning controller changes both effects simultaneously, an apparent coordination gain can instead come from additional total inertia or a favorable training seed. This paper separates sustained common-frequency restoration, common transient support, and differential inertia redistribution. A constrained droop–PI storage layer and a 3 s common-inertia pulse are frozen first; a parameter-shared multi-agent policy can then execute only a bounded, slew-limited, hard-zero-sum two-area residual. On a modified Kundur four-VSG system, 24 paired 60 Hz simulations show that the classical slow and fast layers independently improve their registered endpoints. An outcome-seeing oracle confirms that differential allocation has attainable margin. The evaluated seed-49 policy reduces synchronization loss by 13.96 % with a paired 95% interval of −20.45 % to −4.71 %; its 7.48 % reduction in early inter-area frequency error has an interval of −14.67 % to 0.59 % and therefore does not clear the prospective co-primary gate. All completion, action, storage, common-response, fast-safety, and tail-risk guards pass. These results support the decomposition, but the tested policy does not establish overall adaptive MARL value above the frozen classical reference.**

***Index Terms*—virtual synchronous generator, multi-agent reinforcement learning, grid-forming converter, virtual inertia, frequency coordination**

I. Introduction

Virtual synchronous generators (VSGs) reproduce swing-like behavior in converter controls and provide configurable virtual inertia and damping [1], [2]. When several VSGs operate in parallel, however, their average frequency and their relative motion are different physical coordinates. Raising all inertias can improve the common transient, whereas changing their spatial allocation can alter inter-unit and inter-area motion. A controller that treats these effects as one score may improve system frequency without demonstrating that the VSGs have been better coordinated.

Classical work addresses complementary parts of this problem. Consensus-based secondary control restores frequency in parallel VSG systems [3], [4], while mutual-damping designs target power oscillations and relative modes [5], [6]. These methods make the need for both restoration and damping clear, but they do not by themselves answer a narrower allocation question: after the required common support is fixed, is a disturbance-dependent differential inertia adjustment still useful? This distinction matters because changing total support is usually an easier route to a lower frequency-error metric than learning where that support should be placed.

Recent learning-based VSG controllers adapt inertia and damping using single-agent SAC [7], MADDPG [8], or multi-agent policies [9], [10]. These studies motivate model-free adaptation under changing operating conditions. Nevertheless, jointly learning several inertia–damping actions can mix aggregate support, differential redistribution, and action effort. Selecting the best checkpoint can also yield an optimistic estimate when training-seed variability and a weak reference are not separated from the physical mechanism being claimed.

Power-system RL therefore needs both structural constraints and explicit evaluation uncertainty. Safety filters and Lyapunov-based methods illustrate how learned controls can be restricted by physical requirements [11], [12]; statistical RL work likewise warns that point estimates alone can reverse conclusions in few-run settings [13]. An inference-time weighted actor ensemble considered in the preliminary digest was not retained: a controlled reconstruction did not isolate value beyond the selected actor.

This paper asks one falsifiable question: after sustained restoration and aggregate transient support are frozen, can MARL improve differential coordination without changing fleet-average inertia or harming the common response? The controller is consequently divided by both timescale and coordinate. A slow droop–PI storage layer restores the common frequency, a short common-inertia pulse supplies aggregate fast support, and a shared memoryless actor votes only on one two-area residual. A deterministic projection enforces equal actions within each area and exact normalized zero-sum action across the four VSGs.

The common/differential and fast/slow decomposition makes the incremental MARL claim separately testable. A rank-one coordination layer converts four local actor outputs into a bounded, slew-limited residual that cannot change aggregate virtual inertia. The prospectively staged study uses a matched 24-disturbance bank, full 60 s physical trajectories, paired bootstrap intervals, and actuator/storage/tail guards. The results separate positive mechanism evidence from a negative policy gate: the classical layers and the differential attainability margin are supported, whereas the evaluated policy passes synchronization but fails the uncertainty gate for early inter-area error. Sections II–IV present the formulation, controller, and protocol; Sections V–VII give the results, limitations, and conclusion.

II. Decoupled Control Problem

A. Common and differential coordinates

At the electromechanical timescale, VSG i is represented by

$$M_i \Delta \dot{f}_i + D_i \Delta f_i = \Delta u_i - \Delta P_i, \tag{1}$$

where M_i and D_i denote virtual inertia and damping, Δu_i is the equivalent disturbance, and ΔP_i includes the network-coupled active-power response. The nonlinear phasor-domain model, rather than (1), is used for every reported trajectory.

For $\Delta \mathbf{f} = [\Delta f_1, \ldots, \Delta f_4]^\mathsf{T}$, define

$$\Delta f_\mathrm{c} = \tfrac{1}{4}\mathbf{1}^\mathsf{T}\Delta \mathbf{f}, \tag{2}$$

$$\Delta f_\mathrm{AB} = \tfrac{1}{2}(\Delta f_1 + \Delta f_2) - \tfrac{1}{2}(\Delta f_3 + \Delta f_4), \tag{3}$$

$$L_\mathrm{sync} = \tfrac{1}{4}\sum_{i=1}^{4}(\Delta f_i - \Delta f_\mathrm{c})^2. \tag{4}$$

The common coordinate measures restoration and aggregate excursions; Δf_AB and L_sync measure differential motion. The control design prevents a learned reduction in (4) from being obtained by simply increasing every M_i.

B. Frozen common layers

The slow layer requests equal active power from four storage devices:

$$P^\star_{i,k} = \Pi_\mathcal{F}\left[K_p(f_0 - f_{\mathrm{c},k}) + K_i \sum_{\tau=0}^{k}(f_0 - f_{\mathrm{c},\tau})\Delta t\right], \tag{5}$$

where $K_p = 2.0$ system-pu/Hz/device, $K_i = 0.2$ system-pu/(Hz s)/device, and $\Pi_\mathcal{F}$ enforces power, ramp, converter-capability, SOC, and energy limits with conditional anti-windup. This layer is responsible for sustained restoration.

The fast common layer is a fixed normalized pulse $c_k = 0.25$ for the first 15 samples (3 s) and zero afterwards. With $M_0 = 200$, $\Delta M_\mathrm{max} = 600$, and fixed $D_i = 100$, the common-only reference is $M_i = M_0 + \Delta M_\mathrm{max} c_k$. The slow layer is evaluated against zero support, and the pulse is then evaluated against the slow-only reference before learning is introduced.

III. Hard-Zero-Sum Shared MARL

A. Executed action

One memoryless actor μ_θ is shared by four agents. Each invocation maps a seven-dimensional local observation to $z_{i,k} \in [-1, 1]$. The execution layer aggregates the four votes as

$$q_k^\mathrm{raw} = \frac{q_\mathrm{max}}{2}\left[\mathrm{mean}(z_{1,k}, z_{2,k}) - \mathrm{mean}(z_{3,k}, z_{4,k})\right], \tag{6}$$

$$q_k = \mathrm{clip}(q_k^\mathrm{raw}, q_{k-1} - \Delta q_\mathrm{max}, q_{k-1} + \Delta q_\mathrm{max}), \tag{7}$$

$$\mathbf{r}_k = q_k[1, 1, -1, -1]^\mathsf{T}, \tag{8}$$

followed by $|q_k| \le q_\mathrm{max}$. Here $q_\mathrm{max} = \Delta q_\mathrm{max} = 0.25$. During the active window,

$$M_{i,k} = M_0 + \Delta M_\mathrm{max}(c_k + r_{i,k}), \qquad D_{i,k} = 100. \tag{9}$$

After 3 s, $c_k = \mathbf{r}_k = \mathbf{0}$. Because the magnitude and slew constraints act on the scalar q_k, within-area equality and $\mathbf{1}^\mathsf{T}\mathbf{r}_k = 0$ cannot be broken by elementwise clipping. The physical mapping is audited separately for finite-precision error.

Each agent observes local frequency deviation, common frequency deviation, local-minus-common deviation, local RoCoF, its previous signed residual, local electrical-power change from reset, and a fixed area sign. Components are physically scaled and clipped to $[-5, 5]$. Thus the actors share parameters but receive different local measurements; execution still requires the small coordination layer in (8) to aggregate their votes.

B. Training objective

Centralized training uses a shared two-layer actor and one pair of centralized TD3 critics [14]. The critics receive the joint observation and the executed scalar q_k/q_max; this follows the centralized-critic principle used in multi-agent actor–critic methods [15]. The team reward is

$$R_k = -\frac{L_{\mathrm{sync},k}}{2(0.05)^2} - \frac{\Delta f^2_{\mathrm{AB},k}}{2(0.05)^2} - 0.01\left(\frac{q_k - q_{k-1}}{0.25}\right)^2. \tag{10}$$

No terminal-frequency, RoCoF, storage, or gate term is hidden in (10); those quantities are independent evaluation guards.

Both actor and critics use 64–64 hidden units and a learning rate of 3×10^{-4}. The discount, target-update factor, replay capacity, batch size, and warm-up are 0.99, 0.005, 10^5, 256, and 512 transitions. Policy noise, clipping noise, exploration noise, and the delayed update period are 0.1, 0.2, 0.1, and 2. Seed 49 is trained for 300 episodes of 15 steps, totaling 4500 completed real-ANDES transitions.

IV. Prospective Experimental Protocol

A. Plant and disturbance bank

Experiments use the ANDES framework [16], with version 2.0.0 recorded in the frozen run provenance, and a modified two-area Kundur system [17] with four VSG proxies. The actuator boundary is specific: each PV+GENCLS VSG proxy is accompanied by an independent grid-following ESD1 storage device at buses 12, 16, 14, and 15. It is not a unified physical GFM-BESS implementation. Each storage device is rated 36 MVA/28 MWh, has a 0.36-system-pu power limit, a one-second external ramp, and SOC limits of 0.2–0.8 from an initial 0.5.

The same 24-case bank covers four disturbance locations, positive and negative signs, and moderate (0.65–0.85 pu), strong (0.95–1.15 pu), and edge (1.35–1.50 pu) magnitudes. Every comparison is scenario-paired. The bank was generated and screened before the classical-controller trajectories, but it had been viewed by the oracle and was used for policy training; R278 is therefore a development-bank result, not unseen generalization. The control interval is 0.2 s. Training uses the 3 s active window, whereas every evaluation trajectory runs for 60 s and reports the simulator's physical 60 Hz frequency.

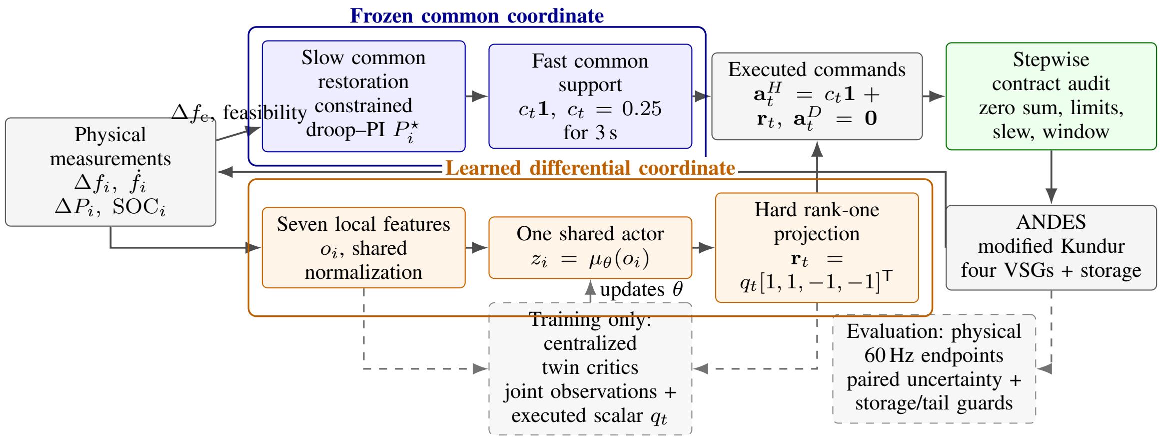

Fig. 1. The proposed controller separates aggregate support from differential coordination. Frozen classical layers regulate the common coordinate; the shared policy can alter only one bounded two-area inertia coordinate whose normalized and physical fleet sums are audited at every step. Dashed paths are used only during centralized training and evidence collection.

B. Stagewise comparisons and gate

Five frozen stages isolate incremental authority: R274 compares slow droop–PI storage with zero requested storage power; R275 adds the common inertia pulse above the slow layer; R276 performs the four-arm zero/slow/fast/combined factorial check; R277 selects, after seeing each outcome, the best of six fixed differential pulses or the reference; and R278 compares the trained policy with the immutable slow-plus-common-pulse reference. R277 is only an attainability upper bound because its selection uses future outcome information.

The R278 co-primary endpoints are the full-trajectory mean of (4) and $\sum_{k=1}^{15} |\Delta f_{\mathrm{AB},k}|\Delta t$. Secondary guards are maximum sampled RoCoF, worst-bus peak, full-horizon common-frequency IAE, and mean common error in the final 10 s. A policy is positive only if both co-primary ratio-of-means effects are at most -2% and both paired 95% interval upper bounds are below zero. Fast metrics may not worsen by more than 5%; slow metrics may not worsen by more than 2%. Completion, action magnitude/slew/window, storage, energy, SOC, saturation, constraint, and empirical CVaR90 guards must also pass.

All intervals use 10 000 shared-index bootstrap resamples of the 24 paired scenarios. The prospective sequence evaluated one pilot seed first and authorized a three-seed continuation only if both co-primary endpoints passed. Consequently, stopping after seed 49 when one endpoint failed is the registered decision, not an omitted experiment.

V. Results and Discussion

A. Classical authority and attainability

The slow layer materially restores the common coordinate. Relative to zero support, it reduces full-horizon mean-frequency IAE by 58.63 % (95% interval −58.67 % to −58.58 %) and final-window common error by 77.29 % (−77.36 % to −77.22 %); all 24 pairs improve on both endpoints. Above this established slow layer, the common-inertia pulse reduces maximum RoCoF by 28.37 %, worst-bus peak by 11.08 %, synchronization loss by 4.29 %, and early inter-area IAE by 9.91 %, with all paired intervals below zero.

The R276 factorial result is *ADDITIVE-ONLY*. None of six registered interaction terms reaches the required -2% material improvement with an interval below zero; the interaction points range from -0.15% to $+1.49\%$ of the zero-support mean. Hence the two classical layers are worth retaining, but their joint value is not evidence of a special coordination synergy.

R277 then asks whether any differential action could still help. Its outcome-seeing oracle chooses a non-reference differential pulse in 18 of 24 cases and reduces synchronization loss by 25.64 % and early inter-area IAE by 19.66 %, with paired intervals wholly below zero. This establishes an attainable differential margin on the viewed bank; it does not establish that a causal policy can identify the action.

TABLE I

R278 SEED-49 EFFECT RELATIVE TO THE FROZEN REFERENCE

Endpoint	Effect (%)	Paired 95% CI (%)	Gate
Synchronization loss	-13.96	[-20.45, -4.71]	pass
Early inter-area IAE	-7.48	[-14.67, +0.59]	fail
Maximum RoCoF	-12.12	[-16.36, -7.50]	pass
Worst-bus peak	-4.54	[-7.51, -1.47]	pass
Mean-frequency IAE	+0.05	[+0.03, +0.08]	pass
Final common error	-0.14	[-0.20, -0.08]	pass

B. Shared-policy pilot

Table I gives the prospective pilot result. The policy reduces synchronization loss by 13.96 %, and its complete interval remains below zero. Early inter-area IAE has a favorable 7.48 % point estimate, but its upper interval bound is $+0.59\%$. Exactly 14 of 24 scenarios improve in synchronization and 12 of 24 improve in early inter-area IAE. The registered rule requires both co-primary endpoints; the classification is therefore `PILOT-NO-GO`, not overall MARL superiority.

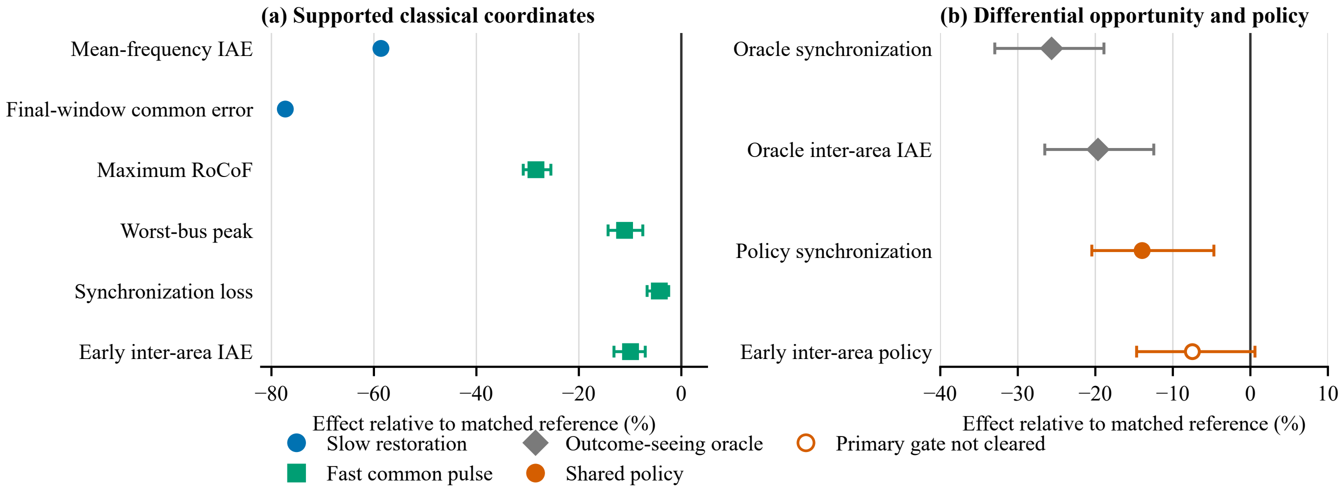

Fig. 2. The classical coordinates and differential attainability margin are supported, while the learned policy clears only synchronization. Points are ratio-of-means effects and bars are paired-bootstrap 95% intervals; negative values favor the left/controller arm. R274 compares slow restoration with zero support, R275 compares the common pulse with the slow layer, and both R277 and R278 compare with the slow-plus-pulse reference. The hollow R278 inter-area point crosses zero and fails the co-primary uncertainty gate.

The negative gate still identifies what the policy did and did not change. RoCoF and peak guards improve, and the full-horizon mean-frequency IAE changes by only $+0.05\%$, well inside the 2% no-harm limit. All 24 policy and reference trajectories complete all 300 samples. Maximum commanded and actual storage power remain below 0.313497 pu, SOC remains in $[0.486074, 0.511454]$, and no saturation reason or constraint violation occurs. Action magnitude, slew, active-window, energy, and all CVaR90 guards also pass. Thus, the hard-zero-sum construction successfully prevents the partial synchronization gain from being purchased through more aggregate inertia or degraded slow restoration.

The policy captures part of the oracle margin but not enough consistent inter-area improvement to justify continuation under the frozen protocol. An action is attainable after observing the outcome, yet a short-horizon memoryless actor trained on the same bank does not reliably select it. The contribution is therefore the separately falsifiable coordination test and its bounded result, rather than a claim that more learning necessarily beats the transparent reference.

VI. Limitations and Reproducibility

R278 contains one training seed and a viewed development bank, so it supports neither seed robustness nor unseen-disturbance generalization. The phasor-domain hybrid actuator is not an EMT validation, unified GFM-BESS, stability certificate, topology study, or deployment result. The original immutable R278 summary was marked invalid because its absolute physical zero-sum threshold, 10^{-8}, was below float32 resolution at the 500-inertia scale. A retained analysis-only amendment set the audit tolerance to four ULP, 1.220703×10^{-4}; the observed maximum was 4.577637×10^{-5}. No trajectory, checkpoint, bootstrap sample, metric, or efficacy threshold changed. These boundaries are necessary for interpreting the positive synchronization signal without extending it beyond the evidence.

VII. Conclusion

This paper separates common restoration, common transient support, and differential inertia redistribution in a four-VSG MARL study. The slow droop–PI storage layer and common-inertia pulse have independently demonstrated authority, and an outcome-seeing oracle confirms that a differential margin exists. A shared TD3 policy restricted to one hard-zero-sum two-area coordinate improves synchronization and respects every registered physical guard, but early inter-area improvement remains uncertain. The prospective two-endpoint gate therefore stops at a no-go for overall adaptive MARL value. The result shows why constrained coordinates, strong matched references, and paired uncertainty are essential: they reveal what learning adds and what it has not yet established.

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